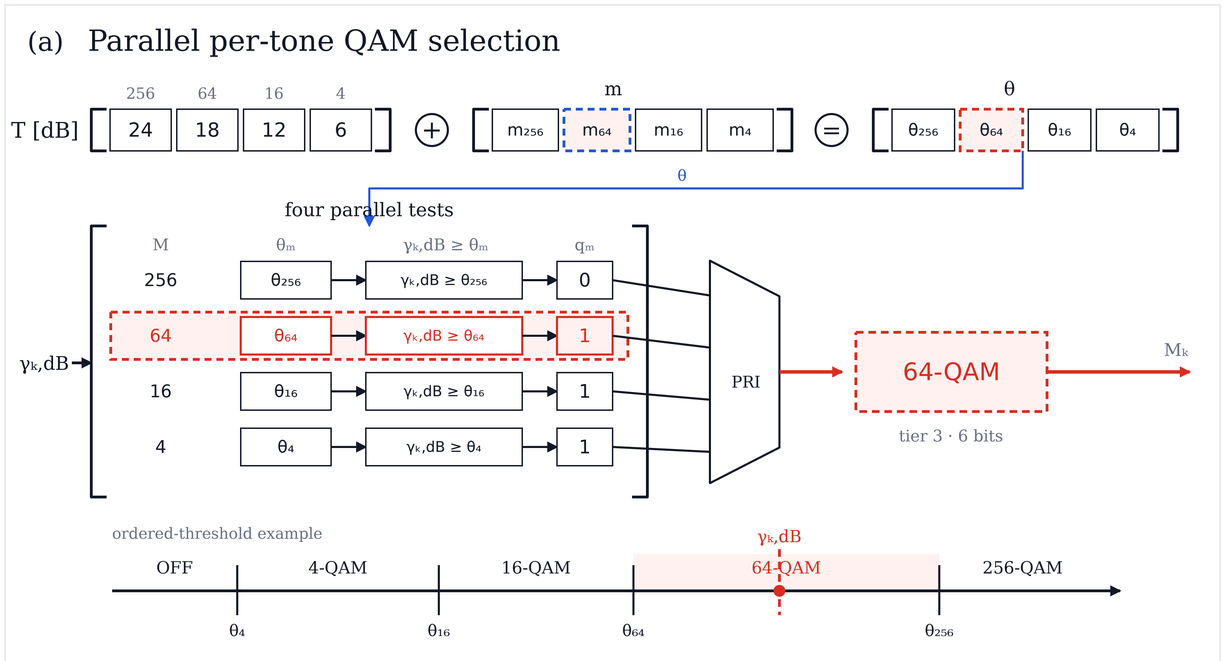
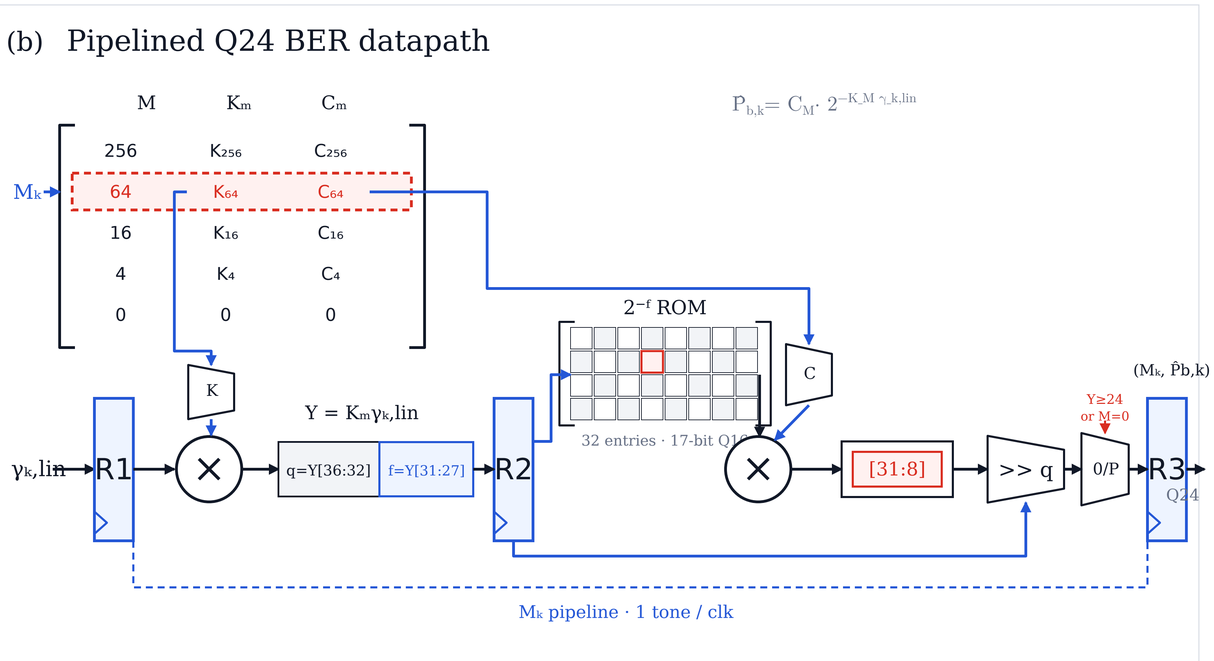
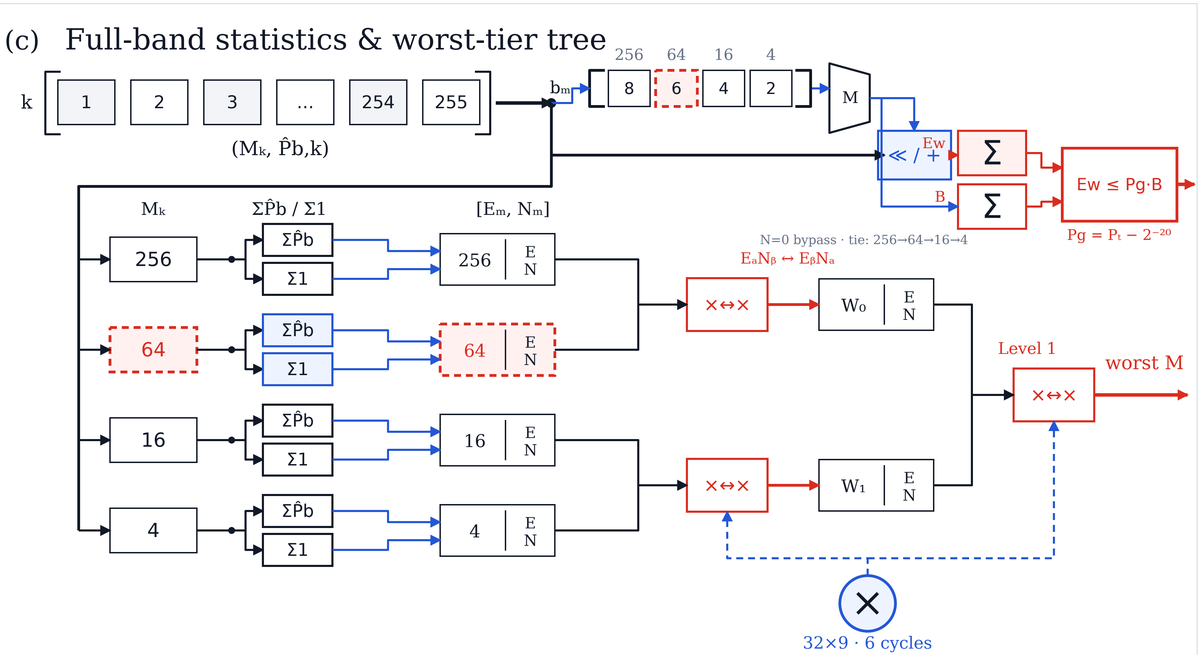
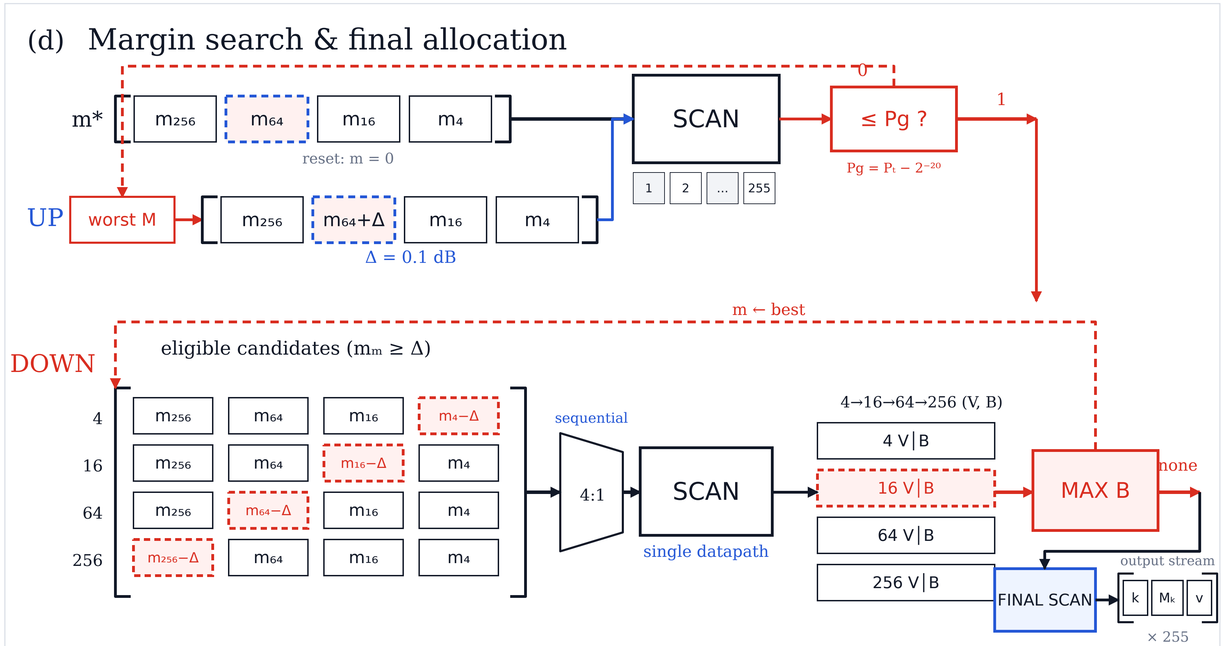

(a) Parallel per-tone QAM selection
256 64 16 4
T [dB]
24 18 12 6
m
m256 m64 m16 m4
θ
θ256 θ64 θ16 θ4
four parallel tests
M θm γk,dB ≥ θm qm
256 θ256 γk,dB ≥ θ256 0
64 θ64 γk,dB ≥ θ64 1
16 θ16 γk,dB ≥ θ16 1
4 θ4 γk,dB ≥ θ4 1
γk,dB
PRI
64-QAM
tier 3 · 6 bits
Mk
ordered-threshold example
γk,dB
OFF
4-QAM
16-QAM
64-QAM
256-QAM
θ4
θ16
θ64
θ256


(b) Pipelined Q24 BER datapath
M Km Cm
256 K256 C256
64 K64 C64
16 K16 C16
4 K4 C4
0 0 0
Mk
Pb,k= CM· 2−K_M γ_k,lin
2−f ROM
32 entries · 17-bit Q16
K
C
Y = Km γk,lin
γk,lin
R1
q=Y[36:32]
f=Y[31:27]
R2
[31:8]
>> q
0/P
R3
Q24
(Mk, P̂b,k)
Y≥24
or M=0
Mk pipeline · 1 tone / clk


(c) Full-band statistics & worst-tier tree
k
1 2 3 … 254 255
(Mk, P̂b,k)
bm
256 64 16 4
8 6 4 2
M
≪ / +
Ew
Σ
B
Ew ≤ Pg·B
Pg = Pt − 2−20
N=0 bypass · tie: 256→64→16→4
EaNβ ↔ EβNa
Mk ΣP̂b / Σ1 [Em, Nm]
256 ΣP̂b Σ1 256 E N
64 ΣP̂b Σ1 64 E N
16 ΣP̂b Σ1 16 E N
4 ΣP̂b Σ1 4 E N
×↔×
W0 E N
W1 E N
Level 1
worst M
32×9 · 6 cycles


(d) Margin search & final allocation
m*
m256 m64 m16 m4
reset: m = 0
SCAN
0
≤ Pg ?
1
1 2 … 255
Pg = Pt − 2−20
UP
worst M
m256 m64+Δ m16 m4
Δ = 0.1 dB
m ← best
DOWN
eligible candidates (mm ≥ Δ)
4
16
64
256
m256 m64 m16 m4−Δ
m256 m64 m16−Δ m4
m256 m64−Δ m16 m4
m256−Δ m64 m16 m4
sequential
4:1
SCAN
single datapath
4→16→64→256 (V, B)
4 V|B
16 V|B
64 V|B
256 V|B
MAX B
none
FINAL SCAN
output stream
k Mk v
× 255